Exam Date & Time: 04-Jan-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH MAKEUP EXAMINATIONS, DEC 2024 - JAN 2025

NETWORK ANALYSIS [BME 2124]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions. Section Duration: 180 mins

1A) Determine the current i_1 in the given circuit Figure Q1a (4)

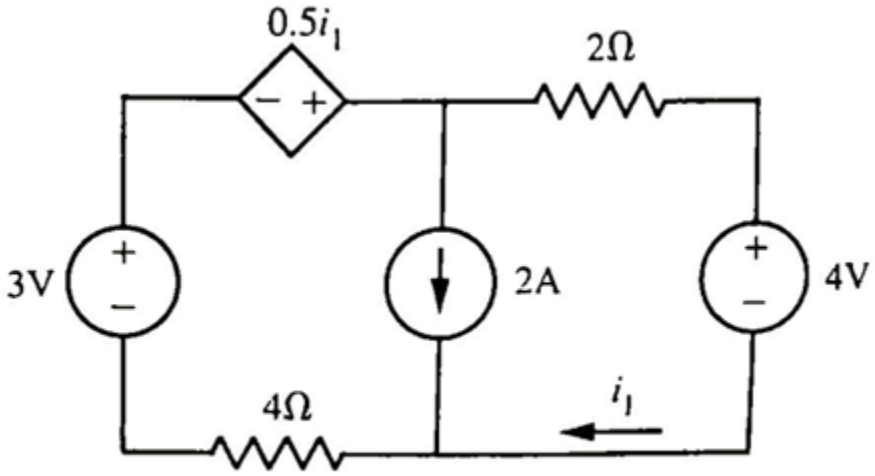


Figure Q1a

1B) Solve for Thevenin's equivalent circuit at the terminals a-b in the given circuit Figure Q1b (3)

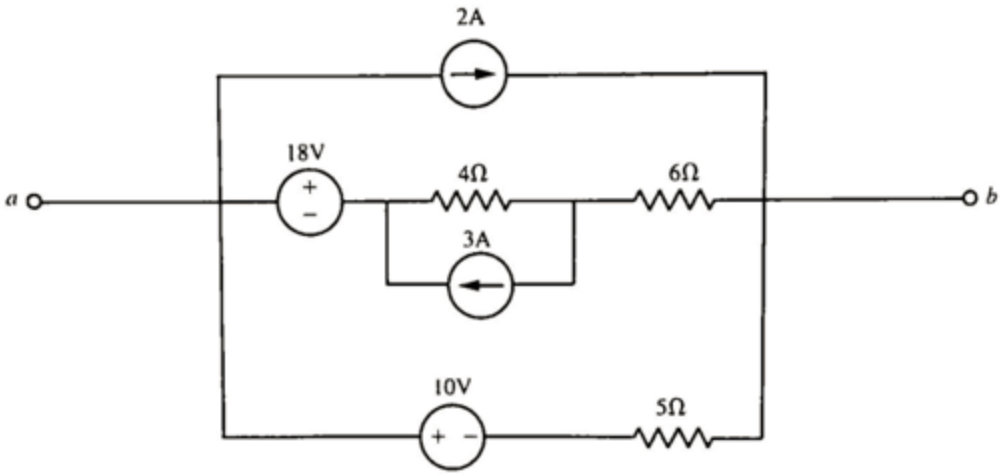


Figure Q1b

1C) In the given circuit Figure Q1c, Evaluate the currents $i_1(0+)$ and $i_L(0+)$. The circuit is in steady state for t=0.

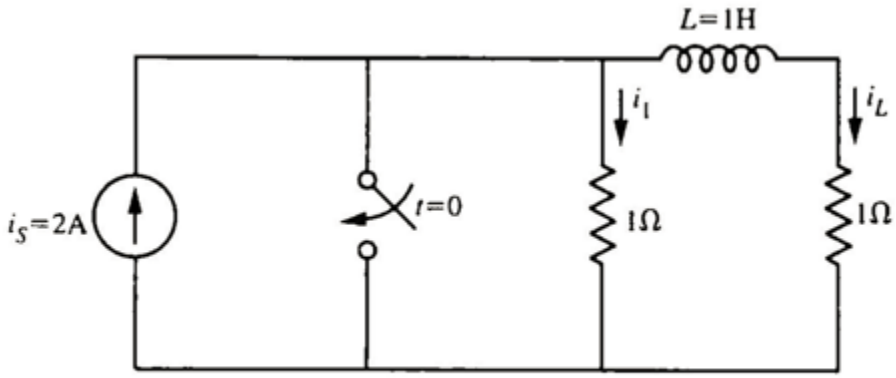


Figure Q1c

(3)

2A) The circuit is in steady state with switch K closed. At t=0, the switch is opened. Calculate the voltage across the switch, v_k and $\frac{dv_k}{dt}$ at t=0+ (5)

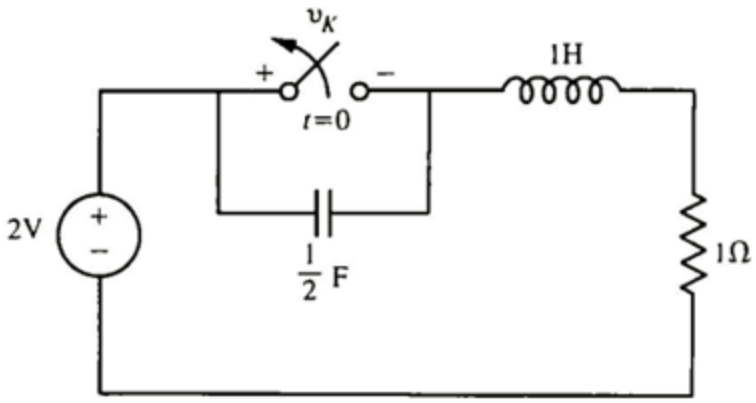


Figure Q2a

2B) A voltage pulse of 10V magnitude and 5μs duration is applied to RC network shown in he Figure Q2b. Evaluate the current i(t) if $R = 10\Omega$ and $C = 0.05\mu F$

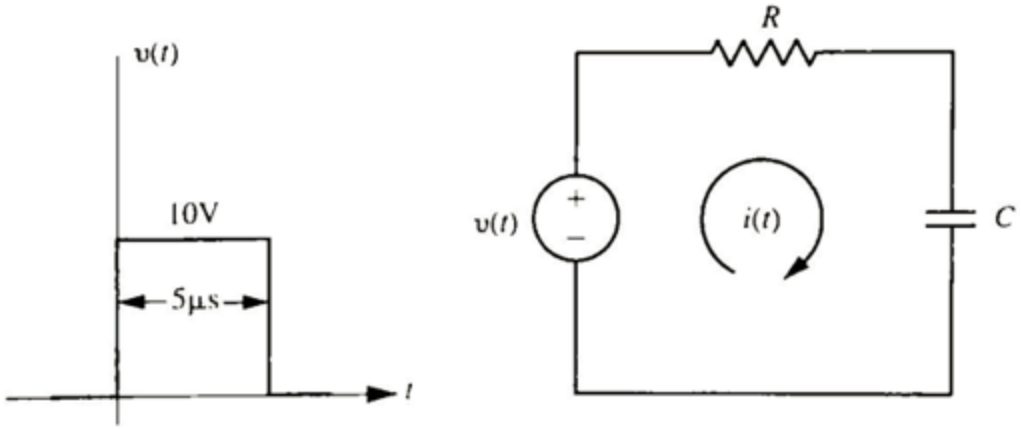


Figure Q2b

(3)

2C) Apply the initial value theorem of $F(s) = \frac{s+1}{(s+1)^2+3^2}$ to obtain its initial value. (2)

3A) Consider the RC circuit shown in Figure Q3a. The input is the rectangular pulse. Compute the current i(t) by assuming circuit is initially relaxed. (4)

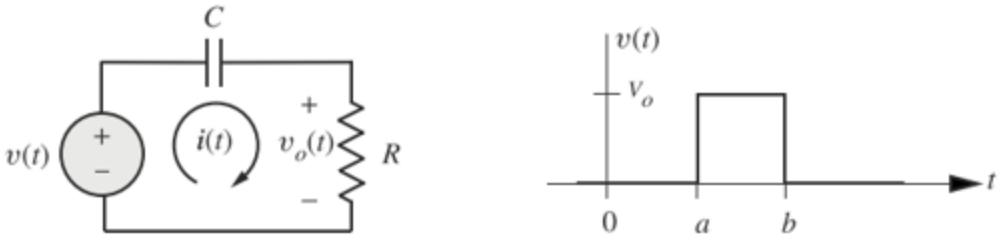
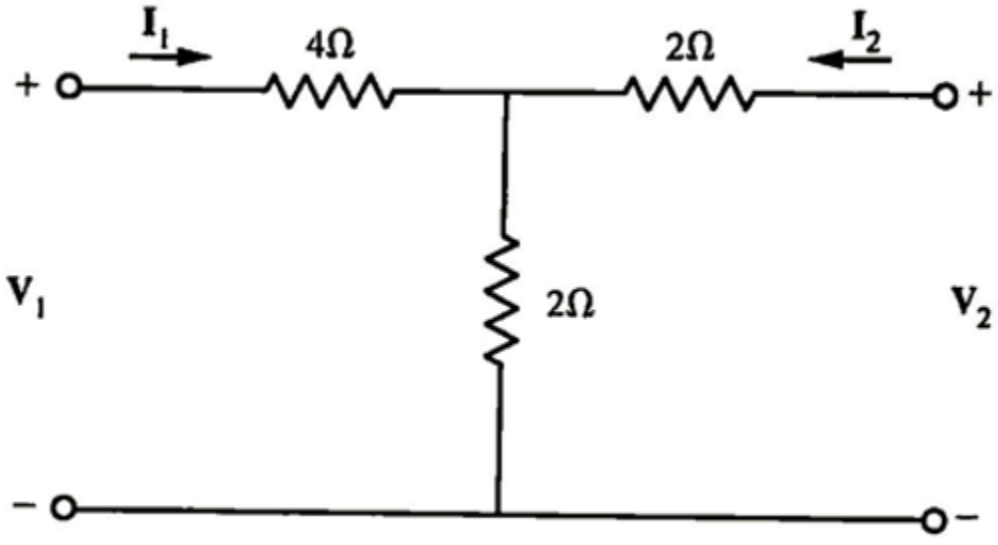


Figure Q3a

3B) Use properties of Laplace transform and solve for $x(t) = \sin^2 \omega_0 t\, u(t)$ (3)

3C) Apply the convolution property to compute output signal from two identical rectangular pulses. Each rectangular pulse has unit amplitude and duration equal to 2T seconds. Also, the pulse is centered at t=T. (3)

4A) Calculate the admittance parameters of the t-network shown in Figure Q4a



(4)

Figure Q4a

4B) Evaluate the Laplace transform of the staircase waveform shown in Figure Q4b. (3)

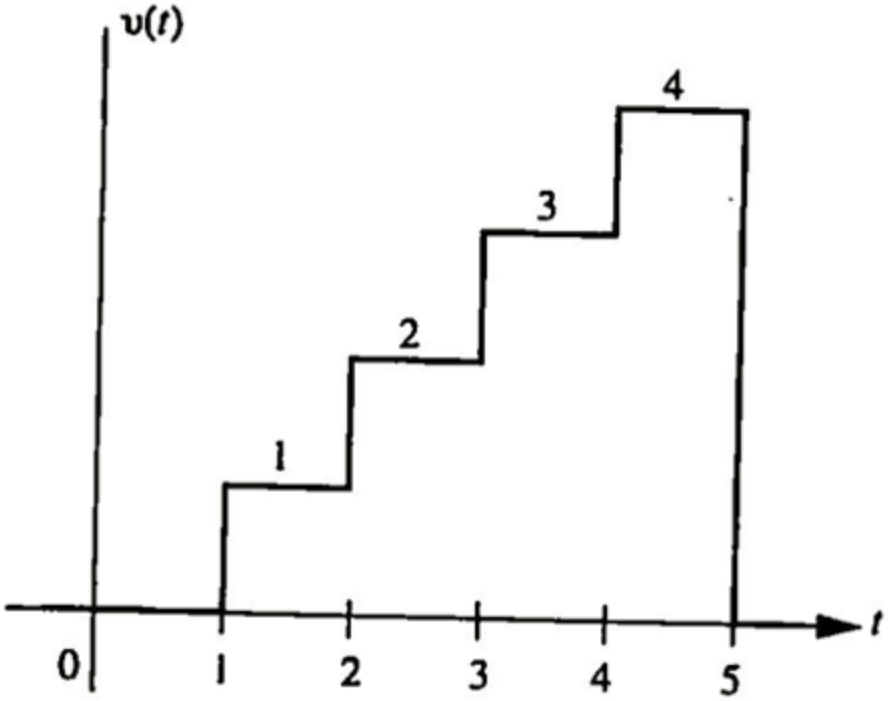
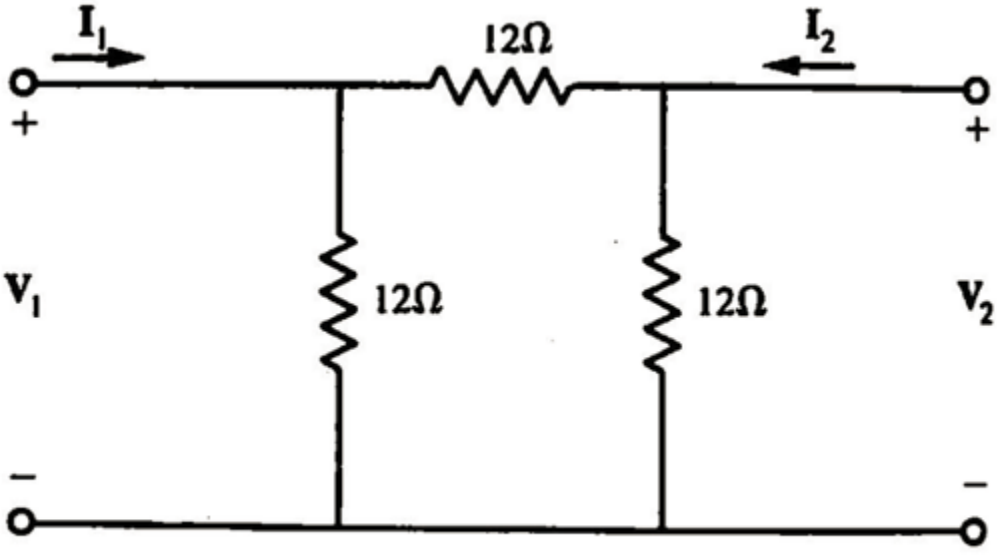


Figure Q4b

4C) Determine the h-parameters of the circuit shown in Figure Q4c



(3)

Figure Q4c

5A) The output $y(t)$ of an LTI system is found to be $e^{-3t}u(t)$ when the input $x(t)$ is $0.5u(t)$ using Laplace transform. (4)

(a) Evaluate the impulse response $h(t)$ of the system.

(b) Analyse the output when the input is $x(t) = e^{-t}u(t)$

5B) Evaluate the Z-parameters of the circuit shown in Figure Q5b. Solve for the current in a 4Ω load if a 24 V source is connected at the input port. (3)

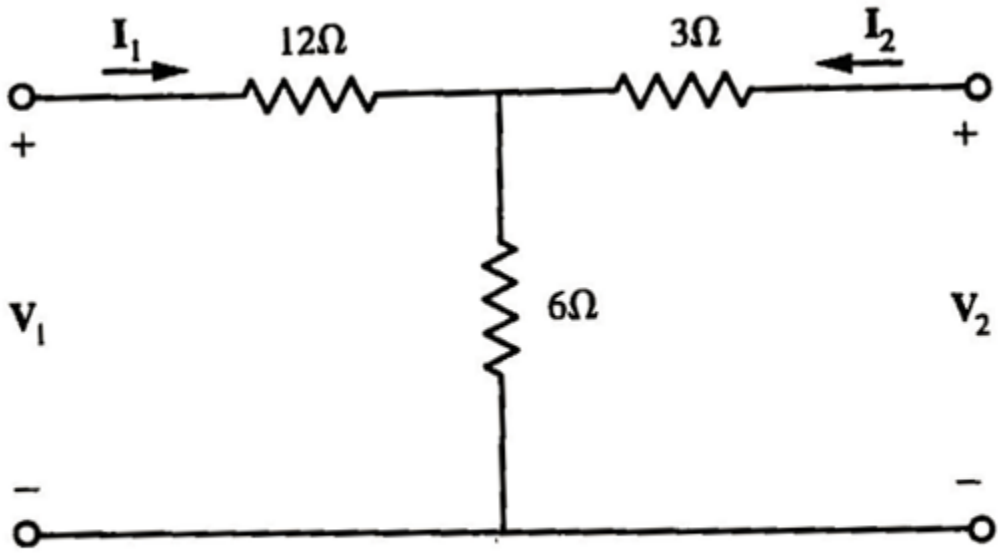


Figure Q5b

5C) Express the current pulse in Figure Q5c in terms of the unit step function. Evaluate the Laplace transform i. $L(i(t))$, ii. $L\left(\int i(t)dt\right)$ (3)

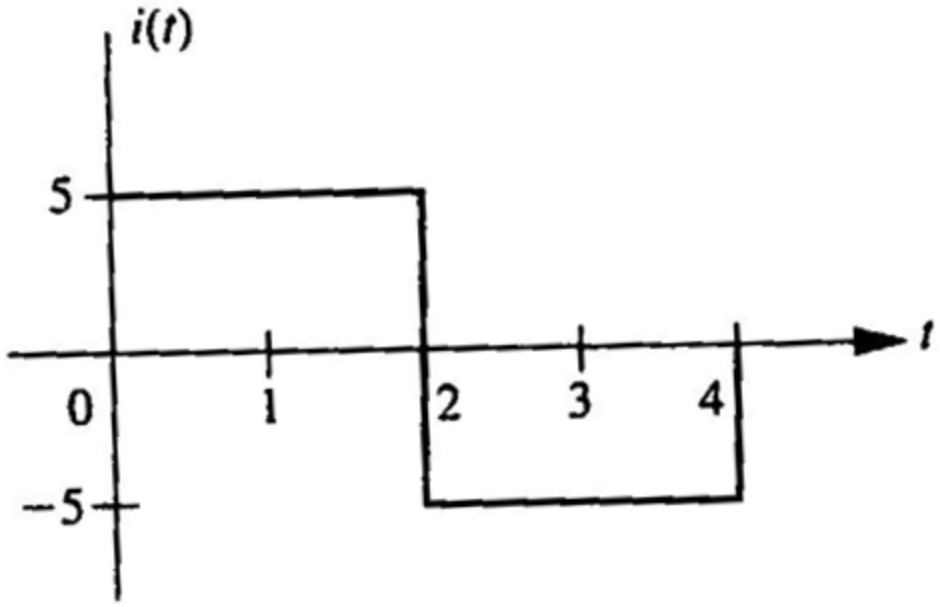


Figure Q5c

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